

Chapter 3 — Finite elements in d dimensions

Global spaces, assembly, integration and error
estimates

MTH3340 · Numerical Methods for PDEs

1 Where we are

We have built, cell by cell, the finite element $(K, \mathcal{V}, \Sigma)$: a polynomial space $\mathcal{V}$ on K , a set Σ of nodal degrees of freedom, and the shape functions ϕ_K^i dual to them — all of it pulled back to one reference cell $\hat{K}$ through Φ_K .

That is step 2 of the recipe. **Step 3 is the one that makes a *space* out of a pile of cells:**

$$\text{cell-local spaces} + \text{continuity constraints} \longrightarrow X_h \subset X.$$

Then we can finally compute: **assemble** the linear system, **integrate** its entries, and ask **how large the error is**.

2 The global finite element space

Definition (Global (conforming) finite element space)

$$X_h \doteq \{v \in X : v|_K \in \mathcal{V} \quad \forall K \in \mathcal{T}_h\}.$$

The hard part is $X_h \subset X$. For problems posed in $H^1(\Omega)$ we need square-integrable gradients, and for piecewise polynomials that means **full continuity**:

$$X_h \subset C^0(\overline{\Omega}) \implies X_h \subset H^1(\Omega).$$

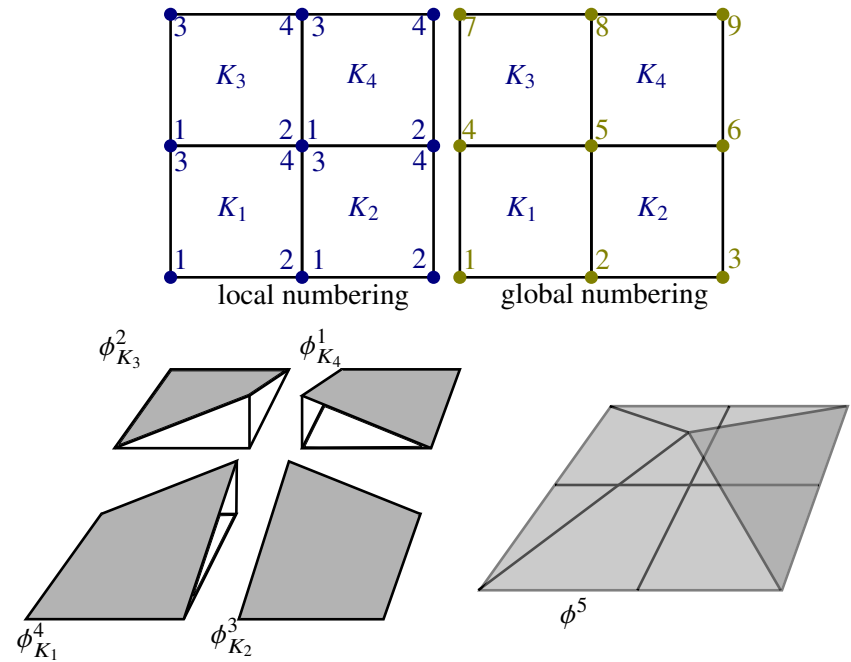
Jumps across a facet would produce a Dirac delta in the distributional gradient — exactly the 1D argument of Chapter 2, one dimension up.

Conformity is what dictates the choice of Σ , and hence which degrees of freedom must be shared between neighbouring cells.

3 Gluing: the local-to-global map

In 1D, two consecutive cells $[x_{i-1}, x_i]$ and $[x_i, x_{i+1}]$ each carry a local node at x_i . Forcing the two local values to agree is exactly **declaring them one global degree of freedom** — and that is how the hat function was born.

Nothing changes in d dimensions except how many cells meet at a node.



The same thing, abstractly

Let $\mathcal{M}_h \doteq \{(i, K) : i \in \mathcal{N}_{\Sigma_K}, K \in \mathcal{T}_h\}$ be all local degrees of freedom of all cells, and declare

$$(i, K) \sim (j, K') \iff \mathbf{x}_i = \mathbf{x}_j :$$

two local degrees of freedom are the same global one *iff their nodes sit at the same point of Ω* .

The global degrees of freedom are the quotient $\mathcal{N}_h \doteq \mathcal{M}_h / \sim$, and $[\cdot]$ is the **local-to-global map**. The global shape functions follow:

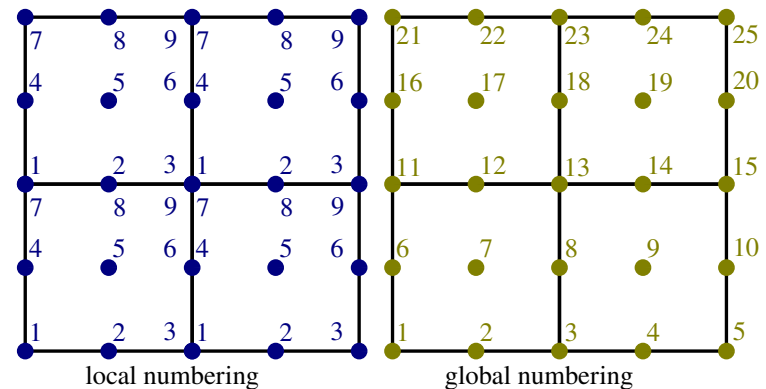
$$\phi^a = \sum_{(i,K) \sim a} \phi_K^i, \quad X_h = \text{span}\{\phi^a\}_{a \in \mathcal{N}_h}.$$

Each ϕ^a picks up *at most one* local shape function per cell, so it is supported on the cells around its node: **locality, hence sparsity**.

Why this gives C^0

Take a facet F shared by K^+ and K^- .

1. $v|_{K^\pm}$ restricted to F is a polynomial **of the same order on F** ;
2. the nodes lying on $\overline{F}$ **determine that polynomial uniquely**;
3. those nodes are **shared**, so both restrictions have the same values there.



Local and global numbering, Q_2 .

Hence the two traces coincide: v is continuous across F . This is where the **conformity of the mesh** is used — F must be a whole facet of both cells.

4 The interpolant

Cell by cell, then glued:

$$\pi_{X_h}(v) \doteq \sum_{K \in \mathcal{T}_h} \pi_K(v) = \sum_{K \in \mathcal{T}_h} \sum_{i \in \mathcal{N}_{\Sigma_K}} \sigma_i(v) \phi_K^i = \sum_{a \in \mathcal{N}_h} v(\mathbf{x}_a) \phi^a.$$

The last equality needs the local values glued to one global degree of freedom to agree, $\sigma_i(v) = \sigma_j(v)$ for $i, j \in [a]$ — true as soon as v is continuous. It is a projection: $\pi_{X_h}(v_h) = v_h$ for $v_h \in X_h$.

A warning that is invisible in 1D. Lagrangian degrees of freedom are *point* evaluations, and for $d > 1$ $H^1(\Omega) \not\subset C^0(\overline{\Omega})$ — an H^1 function need not have nodal values at all. So π_{X_h} is *not* defined on H^1 ; we use it on $\tilde{X} \doteq C^0(\overline{\Omega})$, or on H^{p+1} when we state error bounds.

The 1D embedding we leaned on in Chapter 2 was a one-dimensional luxury.

5 The linear system

Write $u_h = \sum_{b \in \mathcal{N}_h} u_b \phi^b$. Testing against each ϕ^a turns the Galerkin problem into

$$\mathbf{A} \vec{\mathbf{u}} = \mathbf{F}, \quad \mathbf{A}_{ab} \doteq a(\phi^b, \phi^a), \quad \mathbf{F}_a \doteq \ell(\phi^a).$$

Since $a(\cdot, \cdot) = \sum_{K \in \mathcal{T}_h} a_K(\cdot, \cdot)$ (just $\int_{\Omega} = \sum_K \int_K$), we never touch a global shape function:

$$\mathbf{A}_{ab} = \sum_{K \in \mathcal{T}_h} \sum_{\substack{(i,K) \in [a] \\ (j,K) \in [b]}} \mathbf{A}_{ij}^K, \quad \mathbf{A}_{ij}^K \doteq a_K(\phi_K^j, \phi_K^i).$$

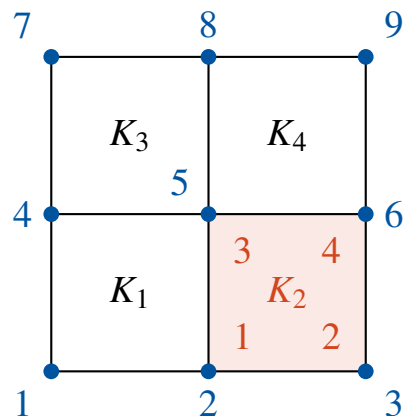
$\mathbf{A}^K$ is the **element matrix**: $n_{\Sigma} \times n_{\Sigma}$, one per cell. **All the work is local; the mesh only supplies indices.**

Assembly in practice

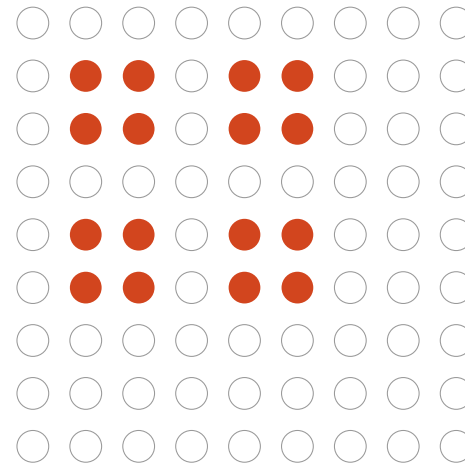
For the cell K_2 below, the local degrees of freedom 1, 2, 3, 4 map to the global ones 2, 3, 5, 6, so the 4×4 element matrix is scattered into those rows and columns:

$$\mathbf{A}[2, 3, 5, 6; 2, 3, 5, 6] \ += \ \mathbf{A}^{K_2}.$$

Scatter-add, one cell at a time. And $\mathbf{A}_{ab} \neq 0$ only if $\text{supp } \phi^a \cap \text{supp } \phi^b$ has positive measure, i.e. only if a and b **share a cell**: the nonzeros per row are bounded independently of $|\mathcal{N}_h|$.



global / local in K_2



$\mathbf{A} \ (9 \times 9)$

6 Numerical integration

Every entry of $\mathbf{A}^K$ is an integral over a physical cell K . Pull it back to $\hat{K}$ and use a quadrature rule there. With $\mathbf{J}_K \doteq \frac{\partial \Phi_K}{\partial \hat{\mathbf{x}}}$ and a rule $Q = \{(\hat{\mathbf{x}}_{\text{gp}}, w_{\text{gp}})\}$ on $\hat{K}$,

$$\int_K \mathcal{I}(\mathbf{x}) \, d\Omega = \int_{\hat{K}} \mathcal{I} \circ \Phi_K(\hat{\mathbf{x}}) |\mathbf{J}_K| \, d\hat{\Omega} \approx \sum_{\hat{\mathbf{x}}_{\text{gp}} \in Q} \mathcal{I} \circ \Phi_K(\hat{\mathbf{x}}_{\text{gp}}) w_{\text{gp}} |\mathbf{J}_K(\hat{\mathbf{x}}_{\text{gp}})|.$$

One reference cell, one quadrature rule, one set of stored evaluations — reused by every cell in the mesh. What is left is to evaluate $\mathcal{I} \circ \Phi_K$, which needs shape functions and their gradients.

Shape functions and gradients at the quadrature points

Values are immediate, since $\phi_K^j \doteq \hat{\phi}^j \circ \Phi_K^{-1}$:

$$\phi_K^j \circ \Phi_K(\hat{\mathbf{x}}_{\text{gp}}) = \hat{\phi}^j(\hat{\mathbf{x}}_{\text{gp}}).$$

Gradients need the chain rule and the inverse function theorem:

$$\nabla \phi_K^j = \nabla(\hat{\phi}^j \circ \Phi_K^{-1}) = \nabla_{\hat{\mathbf{x}}} \hat{\phi}^j \cdot \nabla \Phi_K^{-1},$$

so that, at a quadrature point,

$$\nabla \phi_K^j \circ \Phi_K = \nabla_{\hat{\mathbf{x}}} \hat{\phi}^j \cdot \mathbf{J}_K^{-1} = \mathbf{J}_K^{-T} \nabla_{\hat{\mathbf{x}}} \hat{\phi}^j.$$

So a cell only ever supplies three numbers-worth of geometry: $\mathbf{J}_K$, $\mathbf{J}_K^{-1}$ and $|\mathbf{J}_K|$ at the quadrature points. The quantities $\hat{\phi}^j(\hat{\mathbf{x}}_{\text{gp}})$ and $\nabla_{\hat{\mathbf{x}}} \hat{\phi}^j(\hat{\mathbf{x}}_{\text{gp}})$ are computed **once, for the whole mesh**.

Two model element matrices

Example (Mass matrix (from a reaction term $\int_{\Omega} uv$))

$$\int_K \phi^i \phi^j \, d\Omega = \sum_{\text{gp}} \hat{\phi}^i(\hat{\mathbf{x}}_{\text{gp}}) \hat{\phi}^j(\hat{\mathbf{x}}_{\text{gp}}) w_{\text{gp}} |\mathbf{J}_K(\hat{\mathbf{x}}_{\text{gp}})|.$$

Example (Stiffness matrix (from $\int_{\Omega} \nabla u \cdot \nabla v$))

$$\int_K \nabla \phi^i \cdot \nabla \phi^j \, d\Omega = \sum_{\text{gp}} [\mathbf{J}_K^{-T} \nabla_{\hat{\mathbf{x}}} \hat{\phi}^i] \cdot [\mathbf{J}_K^{-T} \nabla_{\hat{\mathbf{x}}} \hat{\phi}^j] w_{\text{gp}} |\mathbf{J}_K|,$$

all evaluated at $\hat{\mathbf{x}}_{\text{gp}}$.

Which quadrature rule?

On d -cubes, take the **tensor product of 1D Gauss–Legendre**: n points per direction integrate polynomials of degree $\leq 2n - 1$ exactly. For a Q_p element on an affine cell the mass integrand has degree $2p$ per direction, so $n = p + 1$ makes the element matrices **exact — there is no quadrature error at all**.

Where the error is not zero. The right-hand side $\int_K f v$ carries general data f ; and on a non-affine cell $|J_K|$ varies while J_K^{-1} is *rational*, so the integrand is not a polynomial and no rule is exact. There a rule of exactness degree q commits $O(h^{q+1+d})$ on one cell, hence $O(h^{q+1})$ over the $\sim h^{-d}$ cells: **keep that below the discretisation error** — and no further.

Simplices. Symmetric rules are tabulated; for arbitrary order use the **Duffy transformation**, which maps the simplex onto a cube so that tensor-product Gauss can be reused.

Neumann terms: integrating on facets

The right-hand side carries $\int_{\Gamma_N} g_N \nu \, dF$. Same idea, one dimension down: a reference facet $\hat{F}$, a map $\Phi_F : \hat{F} \rightarrow F$, and

$$\int_F \mathcal{I}(\mathbf{x}) \, dF = \int_{\hat{F}} \mathcal{I} \circ \Phi_F(\hat{\mathbf{x}}) |\mathbf{J}_F| \, d\hat{F} \approx \sum_{\text{gp}} \mathcal{I} \circ \Phi_F(\hat{\mathbf{x}}_{\text{gp}}) w_{\text{gp}} |\mathbf{J}_F(\hat{\mathbf{x}}_{\text{gp}})|.$$

Now $\mathbf{J}_F \in \mathbb{R}^{d \times (d-1)}$ is not square, so $|\mathbf{J}_F|$ is a measure, not a determinant:

$$|\mathbf{J}_F| = \left\| \frac{d\Phi_F}{d\hat{\mathbf{x}}} \right\|_2 \quad (d=2), \quad |\mathbf{J}_F| = \left\| \frac{\partial \Phi_F}{\partial \hat{x}_1} \times \frac{\partial \Phi_F}{\partial \hat{x}_2} \right\|_2 \quad (d=3).$$

The facet map is built as before, from the nodes lying on $\overline{\hat{F}}$: $\Phi_F(\hat{\mathbf{x}}) = \sum_{i \in \overline{\hat{F}}} \hat{\phi}_1^i(\hat{\mathbf{x}}) \mathbf{x}^i$.

7 Vector fields and several unknowns

Vector fields. Take $\mathcal{V} \doteq [Q_p]^d$ (or $[\mathcal{P}_p]^d$) with degrees of freedom

$$\sigma_i^\alpha(\mathbf{u}) \doteq \mathbf{u}_\alpha(\mathbf{x}_i), \quad \alpha = 1, \dots, d.$$

Two local degrees of freedom are glued iff **same node and same component**. Globally, with $X_h = \text{span}\{\phi^1, \dots, \phi^n\}$,

$$[X_h]^d = \text{span}\{(\phi^a, 0)^T, (0, \phi^a)^T\}.$$

Several fields. For $(\mathbf{u}, p)$ — Stokes, Navier–Stokes, any multi-physics problem — simply take the **Cartesian product**

$$X_h^1 \times \dots \times X_h^n,$$

gluing degrees of freedom *within* each field only. Different fields may use different orders; that choice is not free (inf–sup), but that is another course.

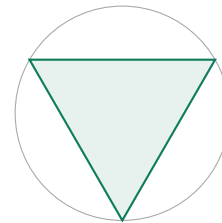
8 Which meshes are we allowed to refine with?

Definition (Shape regular and quasi-uniform families)

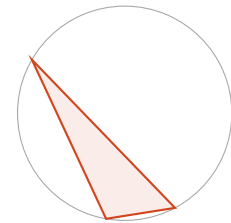
$\{\mathcal{T}_N\}$ is *shape regular* if $\max_{\tau \in \mathcal{T}_N} \frac{\text{diam}(\tau)^d}{|\tau|} \leq c$ for all N , and *quasi-uniform* if additionally $\frac{\max_{\tau} |\tau|}{\min_{\tau} |\tau|} \leq \rho$ for all N . Then $h_{\tau} \doteq |\tau|^{1/d}$ and $h_N \doteq \max_{\tau} h_{\tau}$.

Shape regularity forbids cells that become arbitrarily flat: bisect a rectangle horizontally forever and the mesh stays conforming, but the cells degenerate and h_{τ} stops meaning anything.

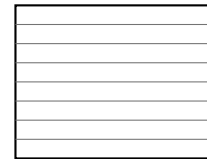
Quasi-uniformity then forbids refining part of the mesh and leaving the rest — it is what makes a *single* h meaningful below.



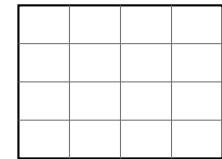
shape regular ✓



sliver ✗



anisotropic ✗



uniform ✓

9 Error bounds

Lemma (Interpolation error in d dimensions)

On a shape-regular, quasi-uniform family with mesh width h , for $\mathcal{Q}_p$ (cubes) or $\mathcal{P}_p$ (simplices) and any $v \in H^{p+1}(\Omega)$,

$$\begin{aligned}\|v - \pi^p(v)\|_{L^2(\Omega)} &\leq Ch^{p+1}|v|_{H^{p+1}(\Omega)}, \\ |v - \pi^p(v)|_{H^1(\Omega)} &\leq Ch^p|v|_{H^{p+1}(\Omega)}.\end{aligned}$$

Exactly the 1D rates. The proof is more technical and out of scope; the *use* is identical. C  a's lemma bounds the error by the best approximation, and the interpolant is one candidate:

$$\|u - u_N\|_{H^1} \leq C \inf_{v_N \in X_h} \|u - v_N\|_{H^1} \leq C \|u - \pi^p(u)\|_{H^1} \leq Ch^p|u|_{H^{p+1}}.$$

In the weaker L^2 norm one gains an order, $\|u - u_N\|_{L^2} \leq Ch^{p+1}|u|_{H^{p+1}}$, by a duality argument we do not prove.

10 The cost of a dimension

The rates are in h ; what we pay for is $N \doteq |\mathcal{N}_h|$. On a quasi-uniform mesh of $\Omega \subset \mathbb{R}^d$ with n cells per direction,

$$N \sim n^d, \quad h \sim \frac{1}{n} = \frac{1}{N^{1/d}}.$$

So $h \sim N^{-1/2}$ in 2D and $h \sim N^{-1/3}$ in 3D, and

$$\|u - u_N\|_{H^1} \lesssim N^{-p/d}.$$

The curse of dimensionality: the same accuracy costs $N^{1/d}$ -fold more work as d grows. This is the argument for raising p rather than refining h — whenever the solution is regular enough to deserve it.

n cells per direction

